

- 15** A small mass undergoes simple harmonic motion about a point O with amplitude x_0 and period T . Its displacement x from O at time $\frac{1}{6}T$ after passing through O is

A $\frac{x_0}{6}$

B $\frac{x_0}{2\sqrt{3}}$

C $\frac{x_0}{2}$

D $\frac{\sqrt{3}x_0}{2}$